

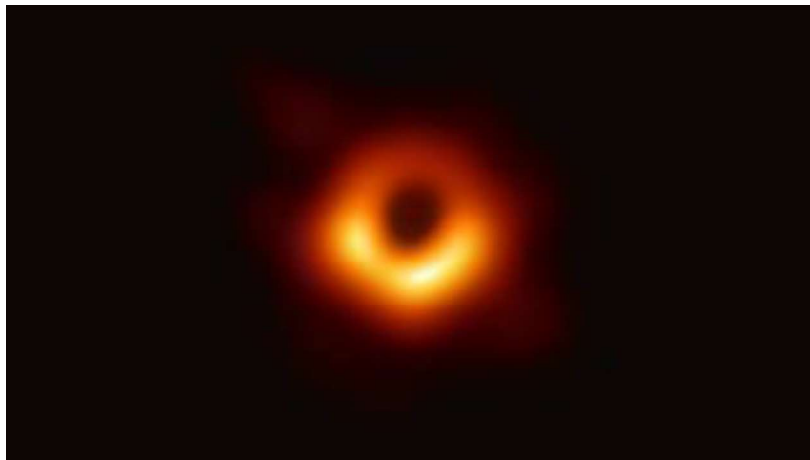
### **1.1.3 Black hole.**

Black hole is an object in the space which is extremely dense and has very strong gravitational attraction not even light can escape it, the milky way possibly contains more than 100 million black holes.

The black hole can be born via two methods the first one black hole is stellar corpse, so they form when a big star dies after the star consume all the fuel, it explodes and leave very compact dense object which create the black hole later on, the other way to form a black hole is the direct collapse of gas, black hole which created this way become much bigger and even reach 100,000 times the size of the sun.

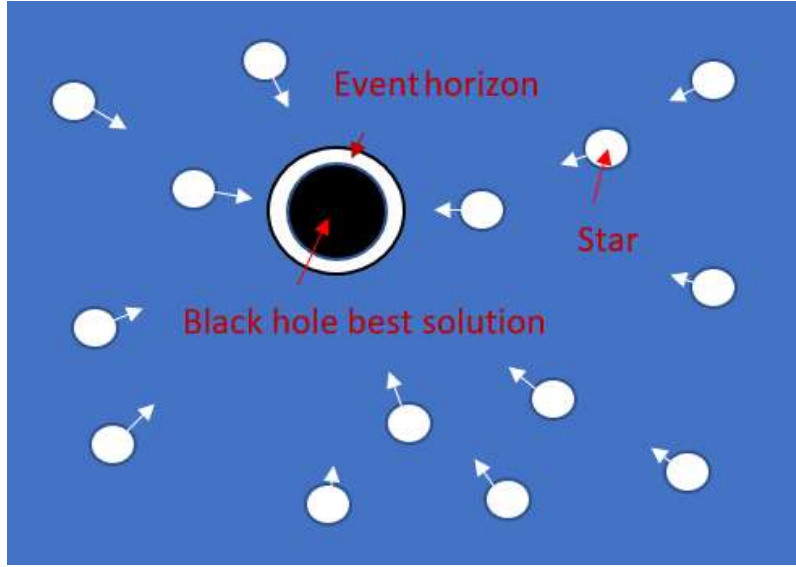
There are three layers for the black hole inner and outer of the event horizon and the singularity. The event horizon of the black hole is the boundary of the black hole where anything passes it even the light cannot escape the black hole anymore and it will be pulled by the black hole because of the constant powerful gravity inside the event horizon, the singularity is the single point in space time where the black hole concentrated.

The black hole can't be seen the same way of the other objects in space. instead detecting the radiation which emit as gas and dust create the shape which can be seen as black hole, one black hole image is shown in figure 2. (Nola Taylor Tillman, 2024)



**Figure 2. Black hole image.**

The black hole algorithm illustration is shown in figure 3, where the best solution become the black hole and star absorb all the other stars to it, if they reach the event horizon the black hole will suck the star and it will be lost, new random star will be generated.



**Figure 3. Black hole algorithm illustration diagram.**

## 1.2 Economic dispatch problem.

Economic dispatch one of the most important optimization related to the power system, the aim is to get the optimal fuel cost for each generating units by getting the set point of each generator power contribution to satisfied the total power demand.

The economic dispatch is difficult problem due to many things like prohibited operating zone, curve of the cost to power generation, and other constraints making the odds to become huge numbers which takes very long time to solve and the more constraint the more complicated the problem of the economic dispatch problem will be. (Nappu, 2024).

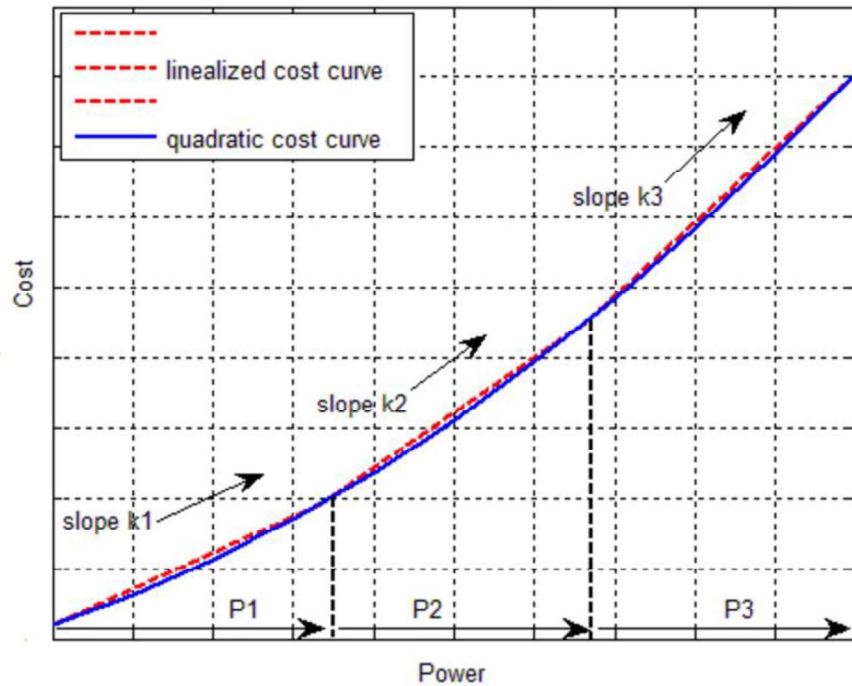
The generators curve normally approximated by cubic or quadratic function as the following equation.

$$C_i(P_i) = a + bP_i + cP_i^2$$

Where  $i$ =the number of the unit,  $C_i$ =cost of the fuel for the unit,  $P_i$ = power generation output of the generator,  $a$ ,  $b$  and  $c$  are fuel cost coefficients of unit  $i$ . the curve is nonlinear which makes it difficult to deal with it the greater the number of units in the system.

Hence finding the optimal solution even without consider other constraint is difficult and time consuming. (M. Zivic Djurovic, 2024)

Figure 4 shows the cost function curve approximation for the generators, which is mostly used while dealing with economic dispatch problem the curve slope is different for each generator unit. (Lombardi, 2024)



**Figure 4. Cost function curve for the generator.**

### 1.3 Project Scope.

The project will be divided into stages to smoothing the implementation of the project, the project will cover the in-depth research of the problem as the first stage moving to other in sequence as the following, project proposal, using black hole algorithm on economic dispatch manually, getting access to Matlab and start learning programming for it, testing the code, comparing the results, modification to the code, getting access to both Matpower and app designer, testing the application of app designer and finalize the project.

Table 1 shows the Gantt chart of the project with allocation of time frame for the takes.

**Table 1. Project Gantt chart.**

| Activities/weeks                                                    | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---------------------------------------------------------------------|---|---|---|---|---|---|---|---|---|
| In-depth research of the problem.                                   |   |   |   |   |   |   |   |   |   |
| Project proposal.                                                   |   |   |   |   |   |   |   |   |   |
| Using black hole algorithm to solve economic load dispatch problem. |   |   |   |   |   |   |   |   |   |
| Learning MATLAB Programming and writing the program.                |   |   |   |   |   |   |   |   |   |
| Testing the program.                                                |   |   |   |   |   |   |   |   |   |
| Comparing and studying the results.                                 |   |   |   |   |   |   |   |   |   |
| Modification to the program.                                        |   |   |   |   |   |   |   |   |   |
| Workshop and reflection report.                                     |   |   |   |   |   |   |   |   |   |
| Report the progress to the supervisor.                              |   |   |   |   |   |   |   |   |   |
| Using the Matpower and app designer.                                |   |   |   |   |   |   |   |   |   |
| Progress report submission.                                         |   |   |   |   |   |   |   |   |   |
| Presentation submission.                                            |   |   |   |   |   |   |   |   |   |
| Poster making and submission.                                       |   |   |   |   |   |   |   |   |   |
| Presentation and demonstration                                      |   |   |   |   |   |   |   |   |   |
| Final report submission                                             |   |   |   |   |   |   |   |   |   |

#### **1.4 Project objectives.**

The objectives for the project as the following.

- Solving economic load dispatch with evolutionary algorithm method and making of application.
- Reducing the emission of carbon dioxide.
- Lowering the cost of generating electrical power.
- Learning more about Matlab.
- Achieving some of kingdom of Bahrain goals.

#### **1.5 Thesis structure.**

- Introduction: includes the background of the project, problem statement, project scope showing different stages process with the Gantt chart and objectives.
- Kingdom of Bahrain perspectives: showing the benefit of the project to kingdom of Bahrain and how it can achieve the needs of kingdom of Bahrain.
- Literature reviews: shows some of the methods used to solve this problem and findings.
- Methodology of the project: black hole algorithm steps, Matlab coding, Matpower and app designer in Matlab.
- Results and discussion: all the results for each stage of the project will be included here.
- Conclusion and recommendation: summary of the project and finding as well as future recommendation to improve the project.

## **2. Bahraini perspectives.**

Kingdom of Bahrain depending almost entirely on fossil fuel mostly natural gas, which affect the global warming issue and making Bahrain vulnerable to fluctuation in global prices and supply. the main goal of Bahrain is to reduce the amount of fossil fuel consumption by finding different ways to reduce the consumption of fossil fuel, some of the actions taken by kingdom of Bahrain to increased low carbon electricity generation using solar energy, Bahrain starts investing more on solar energy to move to green energy. (lowcarbonpower, 2024)

Bahrain has the potential to become the center for the AI development in the region, according to some of the top expert in the field, AI is one of the most talked about technology in the recent years, and it can change the life styles of people to better future this is why development of AI in Bahrain is very important. (zawya, 2024)

Moving to green energy sure will take a while to be achieved during the transmission period anything reduce the amount of fossil fuel used will help Bahrain to achieve the main goal for providing the need of power.

This project focus on reducing the uses of fossil fuel by provided very close to optimal solution to generate the required power and hence reduce the total cost of the power production, moreover reduce in carbon dioxide emission to help with the global warming issue,

There is no extra material also needed for the project other than what it is already being used normally in Bahrain, so the project will not increase anything which affect the environment in anyway, in addition this project is consider as an AI project which contribute to the development of the AI in Bahrain overall.

### **3. Literature reviews.**

Electrical energy plays very important role of our life, and demand for more electrical power is increase every day , learning about different type of power generating is very important, but since most of the electrical energy is provided from fossil fuel right now it is very important to optimize the consumption of fossil fuel and reducing the cost, the economic load dispatch problem is very important topic in the power system, there are different types of approach to solve the economic load dispatch problem based on the methodology, calculation gives the optimum solution, evolutionary algorithm which are based on natural phenomena and many other types.

The need of these algorithm because of the time and complications of the economic load dispatch problem calculation where saving time is considered as cost too. The main advantages of evolutionary algorithm are the simplicity of use even if there is more constrain like increased of number of the units in the system, transmission losses and many other constrains. Comparing different type of optimization study gives general information about the difficulties and important of the economic dispatch problem as well to advantages and disadvantages of some methodology.

Dynamic economic dispatch is differed from the static economic dispatch by incorporating the generator ramp rate constraints, the review discuss two type of the formulation for the dynamic dispatch problem, the first one is the optimal control dynamic dispatch (OCDD), the power system generation modeled as a control system the optimization done for the optimal control with the respect to the ramp rates as input variables. The other formulation is the dynamic economic dispatch (DED), where the optimization is done with respect to dispatchable powers of the generation units in the network. Using of artificial intelligence (AI) as well to hybrid methods are used to solve more complicated objectives or constrains. (X. Xia, 2024)

The methods used are classified into AI and mathematical programming-based methods, for the mathematical programming-based example like lambda iterative method.

It is very important to understand the different between the static and dynamic economic dispatch since complication comes mostly from the dynamic economic dispatch because of the cost to power generated curve slope.

One of the important objectives is the ability of the methods to deal with more constraints if needed without making it very complicated to use.

Differential evolution is smart search algorithm that provide solutions repeatedly to find near the best optimal solution, it is taken from how the evolution operates, the difference for the genetic algorithms (GA) and differential evolutionary algorithms is in genetic algorithm each new generation is related to the solutions before them by using of crossover and mutation and consider one solution represented as binary or integer strings from each group and the new offspring is created from these solutions while the differential evolutionary algorithms combine a vector to generate trial vector which results in more update for the population comparing with the trial vector. (Silesi, 2024) It is very good to understand that changing the population of the solutions more frequently yield better results. Mutation and crossover mean the offspring will always depend on the old generation which is a downside, if there is a better way to get completely new solution will yield better approach for the problem.

Evolutionary algorithms belong to a class of algorithms are designed so as to mimic a behavior of genetic and they are not constrained to genetic they can be inspired by the natural behavior's animals or even phenomena, ant colony algorithm bases on the ant behavior and communication to find the fastest path for the food.

Ant colony optimization depending on the observation of the movement for the ants from their nests to search for food in the shorts path possible, ants communicate by their pheromone, at the start the ants move in random direction to find food the shortest path will have more concentration of pheromone due to more ants choosing this path while the other path pheromone evaporate with time since a smaller number of ants use it. (geeksforgeeks, 2024)

This method is very good since the number of populations can be selected easily, and possibility of completely new solution is generated, the only problem is with the implementation of the algorithm it is a bit difficult to implement.

Black hole algorithm is based on black hole phenomena, black hole got event horizon when anything entre the event horizon it can't escape the black hole and it will be lost forever, the solutions of the algorithm are represented by stars and the best star becomes the black hole, two main steps happen inside the black hole algorithms stars absorption

and stars sucking, the black hole which is the best solution start pull all the other stars to the black hole any star cross the event horizon will be lost and new stars will be created. (zakareya hasan, 2024).

Many other types of algorithms can be used, considering the curve slope of the generator's units, effectiveness of the methods use and complication of implementation of the algorithm, one of the algorithms need to be used, the best type of algorithm to be used are evolutionary algorithms because they are easier to use and give good results.

## 4. Methodology.

The black hole algorithm was chosen for its effectiveness and easy of implementations as well all the information needed is available.

### 4.1 Black hole algorithm heuristic.

- 1- Initialization.
- 2- Generate a random set of solutions.
- 3- Calculate the fitness of each solution.
- 4- Save the optimal solution.
- 5- While not (termination criterion) do
- 6- Execute the algorithm.
- 7- Calculate the fitness of each solution
- 8- Save the optimal solution.
- 9- End while
- 10- Return.

The code steps for the black hole algorithm are as the following.

- 1- Initialize a random population of stars ( $x_i$  and  $i \in [1, N]$ ). Using  $x_i = x_{min} + (x_{max} - x_{min}) \times r$  where  $r$  is random number between (0,1).
- 2- Calculate the fitness of each star depending on the fitness formula.
- 3-  $x_{BH} \leftarrow \text{optimal solution}$ .
- 4- **While** not (termination criterion) **do**
- 5-     **For** each individual  $x_i$  and  $i \in [1, N]$  **do**
- 6-         Star absorption  $x_i^{t+1} \leftarrow x_i^t + (x_{BH} - x_i^t) \times r$  where  $r$  is random number between (0,1).
- 7-     **end for**.
- 8-     Finding the radius  $R \leftarrow \frac{f_{BH}}{\sum_{i=1}^N f_i}$ .
- 9-     **for** each individual  $x_i$  and  $i \in [1, N]$  **do**.
- 10-         Finding the distance  $D_i \leftarrow \sqrt{(x_{1BH} - x_{1i})^2 + \dots (x_{mBH} - x_{mi})^2}$ .
- 11-         **If**  $D_i < R$  **then**.
- 12-              $x_i \leftarrow$  randomly generated individual.
- 13-         **end if**
- 14-     **end for**
- 15-     Calculate the fitness of each star
- 16-      $x_{BH} \leftarrow \text{optimal solution}$ .
- 17- **end while**.
- 18- **return**.

These are the steps to implement the black hole algorithm. (zakareya hasan, 2024)

## 4.2 Black hole algorithm example 1.

Assuming the fitness equation of the system in the example.

$$f = x_1^2 + x_2^2$$

Where  $1 < x_1 < 3$

$0 < x_2 < 5$

This is very easy fitness equation and it is easy to tell the expected final result for  $x_1=1$  and for  $x_2=0$ .

The idea from using such easy fitness equation to simplified the black hole algorithm calculation and compare the results directly with the known final results.

1- Initialization.

Number of stars population = 5

Number of iterations = 3

2- Generate random stars. Using  $x_i = x_{min} + (x_{max} - x_{min}) \times r$

$r$  is random number using the calculator function RAN#.

Star1:  $x_1 = 1 + (3 - 1) \times 0.1 = 1.2$

$x_2 = 0 + (5 - 0) \times 0.03 = 0.15$

Star2:  $x_1 = 1.03$

$x_2 = 2.06$

Star....

3- Fitness

$$f_{star1} = 1.2^2 + 0.15^2 = 1.46$$

The best fitness will be the black hole and red color in the table.

**Table 2. Random stars and star absorption table.**

| X1   | X2   | f     | Star absorption                                                                       | X1   | X2   |
|------|------|-------|---------------------------------------------------------------------------------------|------|------|
| 1.2  | 0.15 | 1.46  |                                                                                       | 1.2  | 0.15 |
| 1.03 | 2.06 | 5.3   | $x_1 = 1.03 + (1.2 - 1.03) \times RAN\#$<br>$x_2 = 2.06 + (0.15 - 2.06) \times RAN\#$ | 1.15 | 0.35 |
| 1.82 | 0.56 | 3.63  | $x_1 = 1.82 + (1.2 - 1.82) \times RAN\#$<br>$x_2 = 0.56 + (0.15 - 0.56) \times RAN\#$ | 1.47 | 0.32 |
| 2.06 | 4.94 | 28.65 | $x_1 = 2.06 + (1.2 - 2.06) \times RAN\#$<br>$x_2 = 4.94 + (0.15 - 4.94) \times RAN\#$ | 1.51 | 0.95 |

|      |      |       |                                                                                       |      |      |
|------|------|-------|---------------------------------------------------------------------------------------|------|------|
| 2.74 | 4.93 | 31.81 | $x_1 = 2.74 + (1.2 - 2.74) \times RAN\#$<br>$x_2 = 4.93 + (0.15 - 4.93) \times RAN\#$ | 2.35 | 1.63 |
|------|------|-------|---------------------------------------------------------------------------------------|------|------|

4- Fitness checking.

From the table again the black hole changed to second star because it has better fitness after the start absorption as shown in the table.

5- Star sucking.

$$R = \frac{1.45}{1.45 + 1.46 + 2.26 + 3.1 + 8.2} = 0.09$$

$$D_{star1} = \sqrt{(1.15 - 1.2)^2 + (0.35 - 0.15)^2} = 0.21$$

**Table 3. Fitness after first absorption and start sucking table.**

| X1   | X2   | F    | R    | D    | X1   | X2   |
|------|------|------|------|------|------|------|
| 1.2  | 0.15 | 1.46 | 0.09 | 0.21 | 1.2  | 0.15 |
| 1.15 | 0.35 | 1.45 | 0.09 | -    | 1.15 | 0.35 |
| 1.47 | 0.32 | 2.26 | 0.09 | 0.32 | 1.47 | 0.32 |
| 1.51 | 0.95 | 3.2  | 0.09 | 0.7  | 1.51 | 0.95 |
| 2.35 | 1.63 | 8.2  | 0.09 | 1.75 | 2.35 | 1.63 |

There is no D less than the R then all the starts move to next iteration.

Note the iteration start after each sucking of the black hole and creating of new star if required.

If any of the distance (D) is less than the radius of the black hole (R) then the same equation which was used to create new star will be used only to replace this star.

As it is shown in table 3 the stars become closer to the black hole after the absorption phase.